PSX Model & Strategy Performance Report

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Cohort separation: the Task 109 general forward cohort's 43 matured outcomes are NOT LLM performance; the LLM cohort has 0 matured outcomes and probability calibration remains forbidden until >=100 genuine matured LLM outcomes exist.

Summary comparison

Model	Status	Promotion
PatchTST	Rejected/unavailable (resource)	no
N-BEATS	Rejected/unavailable (resource)	no
TimesNet	Rejected/unavailable (resource)	no
Informer	Rejected/unavailable (resource)	no
Chronos (Amazon)	Rejected/unavailable (dependency)	no
Lag-Llama	Rejected/unavailable (dependency)	no
Moirai (Salesforce)	Rejected/unavailable (dependency)	no
Time-LLM	Rejected/unavailable (dependency)	no
XGBoost	Trained artifact only — no reliable OOS edge evidence	no
LightGBM	Trained artifact only — no reliable OOS edge evidence	no
CatBoost	Trained artifact only — no reliable OOS edge evidence	no
Random Forest	Trained artifact only — no reliable OOS edge evidence	no
LSTM	Rejected/unavailable (resource)	no
CNN-LSTM	Rejected/unavailable (resource)	no
Transformer-LSTM	Rejected/unavailable (resource)	no
TabNet	Rejected/unavailable (resource)	no
FinGPT	Rejected/unavailable (resource)	no
ARIMA	Rejected/unavailable (data)	no
Holt-Winters	Rejected/unavailable (data)	no
GARCH	Rejected/unavailable (data)	no
Ridge Regression	Trained artifact only — no reliable OOS edge evidence	no
Kalman Filter	Rejected/unavailable (data)	no
Prophet	Rejected/unavailable (dependency)	no
GRU	Rejected/unavailable (resource)	no
XGBoost (CL)	Duplicate alias — counted once	no
LightGBM (CL)	Duplicate alias — counted once	no
Random Forest (CL)	Duplicate alias — counted once	no
LSTM (CL)	Rejected/unavailable (resource)	no
LightGBM-A (CL-2)	Rejected/unavailable (data)	no

LightGBM-B (CL-2)	Rejected/unavailable (data)	no
HistGBM (CL-2)	Rejected/unavailable (data)	no
ExtraTrees (CL-2)	Rejected/unavailable (data)	no
RandomForest (CL-2)	Rejected/unavailable (data)	no
Logistic (CL-2)	Rejected/unavailable (data)	no
MLP (CL-2)	Rejected/unavailable (data)	no
Linear Regression (PI)	Rejected/unavailable (data)	no
Logistic Regression (PI)	Rejected/unavailable (data)	no
Pattern Matcher KNN (PI)	Rejected/unavailable (data)	no
Neural Network (PI)	Rejected/unavailable (data)	no
Technical Engine (GM)	Rule strategy — not ML	no
Announcements Engine (GM)	Rule strategy — not ML	no
Fundamental Engine (GM)	Rule strategy — not ML	no
Macro Engine (GM)	Rule strategy — not ML	no
LightGBM (CG)	Rejected/unavailable (data)	no
XGBoost (CG)	Rejected/unavailable (data)	no
Trend Multi-TF (X)	Rule strategy — not ML	no
Breakout RelVol (X)	Rule strategy — not ML	no
Trend Pullback (X)	Rule strategy — not ML	no
RSI Divergence (X)	Rule strategy — not ML	no
BB Mean Reversion (X)	Rule strategy — not ML	no
Vol Contraction (X)	Rule strategy — not ML	no
Relative Strength (X)	Rule strategy — not ML	no
Market Regime (X)	Rule strategy — not ML	no
Meta-Label CG (X)	Rejected/unavailable (data)	no
Triple Barrier Dyn (X)	Rule strategy — not ML	no
Gian-Ensemble (GIAN)	Rule strategy — not ML	no
Gian-Extender (GIAN-EXT)	Rule strategy — not ML	no
M-Ensemble (M)	Rule strategy — not ML	no
XGBoost	Research only (no demonstrated reliable positive net edge)	no
LightGBM	Research only (no demonstrated reliable positive net edge)	no
CatBoost	Research only (no demonstrated reliable positive net edge)	no
Random Forest	Research only (no demonstrated reliable positive net edge)	no
Ridge Regression	Research only (no demonstrated reliable positive net edge)	no
ExtraTrees	Research only (no demonstrated reliable positive net edge)	no
HistGradientBoosting	Research only (no demonstrated reliable positive net edge)	no
ElasticNet	Research only (no demonstrated reliable positive net edge)	no
HuberRegressor	Research only (no demonstrated reliable positive net edge)	no

ExtraTrees	Research only (superseded by Task 100 evaluation)	no
HistGradientBoosting	Research only (superseded by Task 100 evaluation)	no
ElasticNet	Research only (superseded by Task 100 evaluation)	no
HuberRegressor	Research only (superseded by Task 100 evaluation)	no
SVR	Research only (superseded by Task 100 evaluation)	no
ARIMA	Research only (OOS-evaluated baselines; no net-edge claim)	no
Prophet	Research only (OOS-evaluated baselines; no net-edge claim)	no
jw_reference_lstm	Research only (reference models; no promotion)	no
jw_reference_cnn_lstm	Research only (reference models; no promotion)	no
psx_cost_aware_trade_classifier	Shadow only (frozen cost-aware shadow filter)	no
psx_net_expectancy_meta_model	Shadow only (frozen cost-aware shadow filter)	no
golden_cross_7_30_v1 (Golden Cross 7/30 SMA)	Research only (negative edge after costs)	no
task104 portfolio strategies (fixed_first_hit family)	Research only (prospective/shadow)	no
volatility-aware overlays (A, B 10/15/20, C)	Research only (predeclared family; 10% target is a hypothesi	no
shared strategySignal()/technical-rule implementation + exis	Existing production research signals (unchanged); never auto	no
GLM-4.6 multi-source LLM analyst (z-ai-web-dev-sdk)	Shadow only (LLM cohort distinct from the general Task 109 c	no

Account-level comparison (PKR 100,000)

Strategy	End balance	ROI %	Max DD %	Trades
golden_cross_baseline	60718.23	-39.282	-87775.26	363
overlay_B_vol_target_10pct	97320.37	-2.68	-23.621	428
overlay_B_vol_target_15pct	92758.4	-7.242	-33.656	428
overlay_B_vol_target_20pct	87418.18	-12.582	-42.54	428
overlay_A_fixed_vol_exposure	79652.78	-20.347	-62.414	428
overlay_C_expansion_protection	79652.78	-20.347	-62.414	428
cash	100000.0	0.0	0.0	0
buy_hold_equal_weight	117499.27	17.499	None	None

Full profiles, per-horizon metrics, cost assumptions and evidence lineage: see docs/reports/psx_model_strategy_performance_report.md and data/reports/model_strategy_performance_latest.json. Missing values are marked 'Not available in source evidence'.